Module 7: Knowledge Graphs for Lung Adenocarcinoma

I built a knowledge graph for lung adenocarcinoma (LUAD) using the target list from Module 2, which drew on CPTAC LUAD proteogenomic data (National Cancer Institute, 2020), protein tractability scores from Module 3, and selected drug, variant, pathway, and phenotype evidence. My Neo4j graph contains 29 nodes and 29 relationships, including my 15 ranked LUAD genes. The goal was to connect these different types of evidence to identify possible biomarkers, understand biological relationships, and explore drug-repurposing hypotheses.

1. Schema and Evidence

I used six node types: Gene, Disease, Phenotype, Pathway, Drug, and Variant. Genes represent LUAD candidates; disease nodes include LUAD and idiopathic pulmonary fibrosis (IPF). Phenotypes include cough and dyspnea. Pathways show functional context, drugs connect to molecular targets, and variants represent genetic changes. I did not add patient nodes because I did not have linked real LUAD patient records.

The 15 LUAD genes came from Module 2, with RNA and protein expression, mutation frequency, multi-omics scores, and Module 3 tractability scores. I left GWAS p-values empty because my data did not include them. I used consistent gene identifiers across datasets and a Neo4j uniqueness constraint to avoid duplicate entities.

Relationship	Meaning
ASSOCIATED_WITH	Gene–disease association
TARGETS	Drug–target relationship
HAS_PHENOTYPE	Disease–symptom relationship
PARTICIPATES_IN	Gene–pathway membership
OCCURS_IN	Variant–gene relationship

Every edge includes a source and confidence property, allowing established evidence to be distinguished from weaker associations. These confidence labels are qualitative and are not experimentally measured probabilities. The final database contained 29 nodes and 29 relationships.

2. Top Targets and Biological Interpretation

PageRank measures a node's importance based on its graph connections. The highest PageRank values among all gene nodes were CYP2C19 (0.0418), EGFR (0.0418), and PDGFRB (0.0286). EGFR is a well-established LUAD driver; CYP2C19 was added for Module 5 drug-metabolism context, not as a LUAD driver; and PDGFRB came from the related pulmonary fibrosis example. Therefore, a high PageRank alone does not establish a strong LUAD biomarker (Cancer Genome Atlas Research Network, 2014).

All 15 original LUAD candidates had the same PageRank (0.0155), because most have the same direct connection to LUAD. I therefore combined normalized rank percentiles using 30% centrality, 42% multi-omics evidence, and 28% druggability. These are transparent exploratory weights, not a clinically validated model.

Rank	Gene	PageRank	Combined score
1	COL6A6	0.0155	0.7667
2	ACTN2	0.0155	0.7480
3	SCN7A	0.0155	0.7480
4	TNR	0.0155	0.6827
5	TNXB	0.0155	0.6080

COL6A6 ranked first because of strong multi-omics evidence, consistent with its high ranking in Module 3. SCN7A was interesting because of its relatively high predicted small-molecule tractability, but that does not prove therapeutic value. ACTN2 and SCN7A have equal displayed combined scores, so their ordering is not a meaningful biological difference. ZNF536 was a non-obvious hit: RNA and protein expression changed in opposite directions and its predicted druggability was low, requiring additional experimental checks.

Community Detection

Greedy modularity community detection identified a group containing COL6A6, COL11A1, and collagen formation (R-HSA-1474290). This may reflect extracellular-matrix remodeling or stromal changes in LUAD. Other groups connected EGFR, its L858R variant, and osimertinib, and MASP1 with lectin complement activation. Reactome provides functional context, but an algorithmic community is not proof of a biological subtype (Reactome, n.d.-a, n.d.-b).

3. A Hypothesis From a Graph Path

The professor's two-hop Cypher query returned:

Osimertinib → EGFR → Lung adenocarcinoma

This is consistent with osimertinib's approved use in eligible EGFR-mutated non-small-cell lung cancer, including certain tumors with EGFR L858R (U.S. Food and Drug Administration, 2024).

The shared-phenotype query returned:

Nintedanib → PDGFRB → Idiopathic pulmonary fibrosis → Cough ← Lung adenocarcinoma

Dyspnea was also shared. Nintedanib inhibits multiple receptor tyrosine kinases, including PDGFR, and is used in fibrosing lung disorders (DailyMed, 2026).

This result suggested asking whether PDGFRB-related signaling matters in LUAD tumors with fibrosis or stromal remodeling. However, cough and dyspnea are common symptoms: sharing them does not prove a shared molecular cause or that nintedanib treats LUAD.

To test the hypothesis, I would compare PDGFRB expression and stromal markers in independent LUAD tumors and normal lung tissue. If the evidence supports pathway activity, I would test nintedanib in LUAD and tumor–fibroblast co-culture models using dose-response assays, target engagement, cell viability, and normal-

cell controls. Knockdown and rescue experiments could assess whether any effects depend on PDGFRB rather than other kinase targets.

4. Limitations, Evidence Quality, and AI Use

A graph edge is evidence, not necessarily biological truth. A gene's changed expression does not prove it drives LUAD. Many gene–disease edges were supported by observational multi-omics data rather than functional studies. The shared-phenotype path is weaker still, because cough and dyspnea occur in many respiratory diseases. The small graph (29 nodes and 29 relationships) also means PageRank and community results depend heavily on which connections were included.

The CYP2C19 variants reflect Module 5 pharmacogenomics, not LUAD oncogenesis. I did not have patient-level LUAD outcomes, so I did not perform patient stratification. More curated pathways, direct functional evidence, well-characterized clinical variants, patient genomics, and treatment outcomes could strengthen or change the ranking and path interpretation.

AI tools can assist with identifier matching, writing Cypher queries, and code interpretation, but their suggestions require independent checking. I compared the gene list with Modules 2 and 3, confirmed the 29-node/29-edge counts, reviewed actual query outputs, and included source and confidence properties on edges. I also compared drug and pathway claims with regulatory and biological sources. These checks reduce errors but cannot replace experimental validation.

Conclusion

This assignment connected my previous LUAD analyses into one searchable graph. It recovered the established osimertinib–EGFR relationship, identified a collagen-related community, and suggested a hypothesis involving PDGFRB signaling in fibrotic lung disease. COL6A6, ACTN2, and SCN7A led my combined biomarker ranking.

The key lesson is that more graph connections do not automatically mean better biomarkers; centrality must be interpreted with molecular evidence and known biology.

References

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